

IPv6 (Part-2)

- Topics covered:
 - IPv6 address configuration (Modified EUI-64)
 - IPv6 address types
 1. Global unicast
 2. Unique local
 3. Link local
 4. Multicast
 5. Others...

IPv6 address configuration (Modified EUI-64)

EUI-64 (Modified EUI-64)

EUI stands for Extended Unique Identifier.

In IPv6, the technically correct term is modified EUI-64, but it is usually just called EUI-64.

For CCNA purposes: “Modified EUI-64” and “EUI-64” mean the same thing.

EUI-64 is a method of converting a 48-bit MAC address into a 64-bit interface identifier.

This interface identifier becomes the host portion of a /64 IPv6 address.

- IPv6 address length: 128 bits
- /64 means:
 - 64 bits = network portion
 - 64 bits = host portion

Converting a MAC Address to an EUI-64 Interface Identifier

Step-by-Step Process

Example MAC address:

1234 5678 90AB

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Step 1: Divide the MAC Address in Half

1234 56 | 78 90AB

Step 2: Insert FFFE in the Middle

1234 56FF FE78 90AB

Step 3: Invert the 7th Bit

- Each hexadecimal digit = 4 bits
- The 7th bit is:
 - The 3rd bit of the second hexadecimal digit

Original:

12 → 0001 0010

^

7th bit = 1

Invert the bit:

0001 0010 → 0001 0000

Resulting hexadecimal change:

1234 56FF FE78 90AB

↓

1034 56FF FE78 90AB

Result-> EUI-64 Interface ID: 1034:56FF:FE78:90AB

This 64-bit value is appended to the network prefix.

How to convert the MAC address:

1: Divide the MAC address in half

1234 5678 90AB → 1234 56 | 78 90AB

2: Insert FFFE in the middle

1234 56FF FE78 90AB

3: Invert the 7th bit

1234 56FF FE78 90AB → 1034 56FF FE78 90AB

0010 → 0000

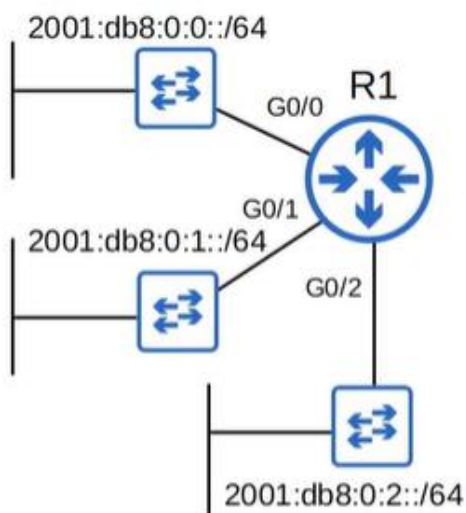
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EUI-64 Practice Conversions

MAC Address	EUI-64 Interface Identifier
782B CBAC 0867	7A2B CBFF FEAC 0867
0200 4C4F 4F50	0000 4CFF FE4F 4F50
0050 56C0 0001	0250 56FF FEC0 0001
00FF 6BA6 F456	02FF 6BFF FEA6 F456
96AB 6D6B 98AE	94AB 6DFF FE6B 98AE

Configuring IPv6 Addresses Using EUI-64

Network Diagram:



Router Configuration (CLI):

```
R1(config)#int g0/0
R1(config-if)#ipv6 address 2001:db8::/64 eui-64
R1(config-if)#no shutdown
R1(config-if)#
R1(config-if)#int g0/1
R1(config-if)#ipv6 address 2001:db8:0:1::/64 eui-64
R1(config-if)#no shutdown
R1(config-if)#
R1(config-if)#int g0/2
R1(config-if)#ipv6 address 2001:db8:0:2::/64 eui-64
R1(config-if)#no shutdown
```

- Use the given prefix
- Automatically generate the host portion using EUI-64

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Verifying MAC Addresses CLI Output:

```
R1#show interfaces g0/0
GigabitEthernet0/0 is administratively down, line protocol is down
Hardware is iGbE, address is 0cf8.2236.8500 (bia 0cf8.2236.8500)

R1#show interfaces g0/1
GigabitEthernet0/1 is administratively down, line protocol is down
Hardware is iGbE, address is 0cf8.2236.8501 (bia 0cf8.2236.8501)

R1#show interfaces g0/2
GigabitEthernet0/2 is administratively down, line protocol is down
Hardware is iGbE, address is 0cf8.2236.8502 (bia 0cf8.2236.8502)
```

Verifying IPv6 Addresses CLI Output:

```
R1(config-if)#do show ipv6 interface brief
GigabitEthernet0/0    [up/up]
FE80::EF8:22FF:FE36:8500
2001:DB8::EF8:22FF:FE36:8500
GigabitEthernet0/1    [up/up]
FE80::EF8:22FF:FE36:8501
2001:DB8:0:1:EF8:22FF:FE36:8501
GigabitEthernet0/2    [up/up]
FE80::EF8:22FF:FE36:8502
2001:DB8:0:2:EF8:22FF:FE36:8502
GigabitEthernet0/3    [administratively down/down]
unassigned
```

- C in MAC becomes E due to 7th-bit inversion
- FFFE inserted in the middle
- Interfaces have very similar IPv6 addresses

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In depth Clarity:

1 MAC → EUI-64 (this part you already know)

Your MAC address (from output):

0c:f8:22:36:85:00

Step-1: Split MAC

0c:f8:22 | 36:85:00

Step-2: Insert ff:fe in the middle

0c:f8:22:ff:fe:36:85:00

Step-3: Flip the 7th bit (Universal/Local bit)

- First byte: 0c
- Binary of 0c = 00001100
- Flip 7th bit → 00001110
- Hex becomes 0e

So final Interface ID is:

0e:f8:22:ff:fe:36:85:00

Written in IPv6 format:

0ef8:22ff:fe36:8500

✓ This part is 100% correct

2 Why does it show 2001:db8::ef8:22ff:fe36:8500 ?

Because EUI-64 does NOT create the full IPv6 address alone.

👉 It only creates the last 64 bits (Interface ID)

👉 The first 64 bits come from the IPv6 prefix

3 You configured this command 🙋

ipv6 address 2001:db8::/64 eui-64

So the router does this automatically:

[Prefix (64 bits)] + [Interface ID (64 bits)]

Prefix:

2001:db8::

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Interface ID:

0ef8:22ff:fe36:8500

Full IPv6 address:

2001:db8::0ef8:22ff:fe36:8500

4 Then why is 0ef8 shown as ef8 ?

Because IPv6 removes leading zeros.

0ef8 → ef8

So:

2001:db8::0ef8:22ff:fe36:8500

becomes:

2001:db8::ef8:22ff:fe36:8500

✓ Both are the SAME address

5 Final clarity (MOST IMPORTANT)

- 0ef8:22ff:fe36:8500
👉 Only the Interface ID
- 2001:db8::ef8:22ff:fe36:8500
👉 Full IPv6 address = Prefix + Interface ID

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Difference B/w Normal Configuration & EUI-64

◆ Method 1: Manual IPv6 address (FULL address given)

Example from your screenshot:

```
R1(config)#
R1(config)#ipv6 unicast-routing
R1(config)#
R1(config)#int g0/0
R1(config-if)#ipv6 address 2001:db8:0:0::1/64
R1(config-if)#no shutdown
R1(config-if)#
R1(config-if)#int g0/1
R1(config-if)#ipv6 address 2001:db8:0:1::1/64
R1(config-if)#no shutdown
R1(config-if)#
R1(config-if)#int g0/2
R1(config-if)#ipv6 address 2001:0db8:0000:0002:0000:0000:0000:0001/64
R1(config-if)#no shutdown
R1(config-if)#
```

What happens here?

- You are manually writing the full IPv6 address
- Interface ID is fixed (::1)
- Router does NOT use MAC address
- Same address every time

Meaning:

Prefix (64 bits) + Manually given Interface ID

Example:

Prefix: 2001:db8:0:0::

Host ID: ::1

- ✓ Simple
- ✗ Not automatic
- ✗ No MAC involvement

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♦ Method 2: EUI-64 (Automatic using MAC address)

Example:

```
R1(config)#int g0/0
R1(config-if)#ipv6 address 2001:db8::/64 eui-64
R1(config-if)#no shutdown
R1(config-if)#
R1(config-if)#int g0/1
R1(config-if)#ipv6 address 2001:db8:0:1::/64 eui-64
R1(config-if)#no shutdown
R1(config-if)#
R1(config-if)#int g0/2
R1(config-if)#ipv6 address 2001:db8:0:2::/64 eui-64
R1(config-if)#no shutdown
```

What happens here?

- You give only the network prefix
- Router automatically creates host part
- Uses MAC address
- Host ID changes if MAC changes

Meaning: Prefix (64 bits) + Interface ID (from MAC)

Result looks like:

2001:db8::ef8:22ff:fe36:8500

- ✓ Automatic
- ✓ Unique per interface
- ✗ Address looks complex

MAIN DIFFERENCE (THIS IS THE KEY)

Feature	Manual Address	EUI-64
Host part	You write it	Router generates
Uses MAC	✗ No	✓ Yes
Address	Simple (::1)	Long & complex
Control	Full control	Automatic
Common use	Labs, gateways	Real networks

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Why the 7th Bit Is Inverted

MAC address types:

- UAA (Universally Administered Address)
- LAA (Locally Administered Address)

The 7th bit is the U/L bit.

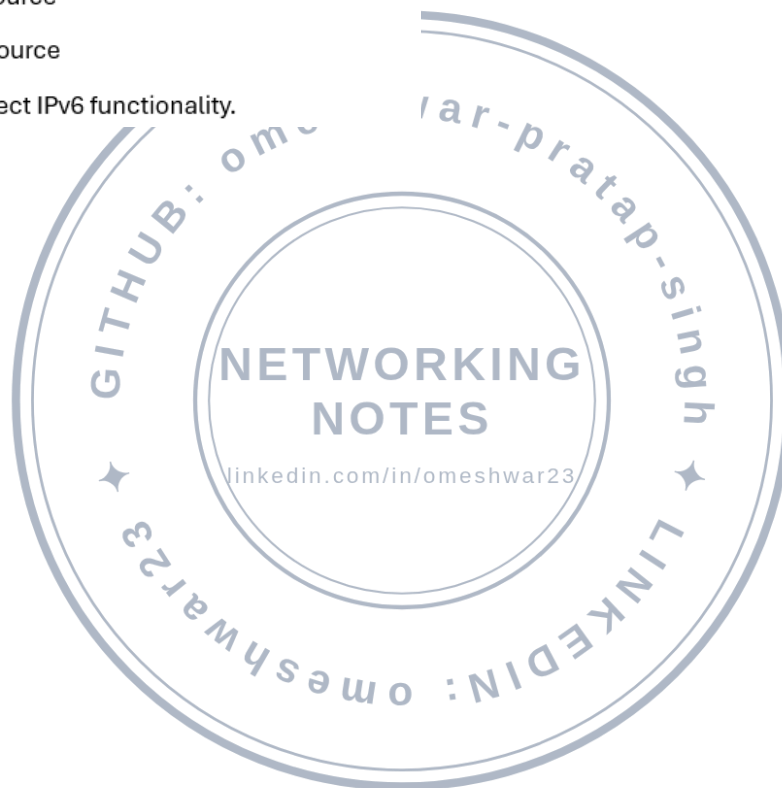
Normal MAC meaning:

- 0 → UAA
- 1 → LAA

In IPv6 EUI-64:

- 0 → LAA source
- 1 → UAA source

This does not affect IPv6 functionality.



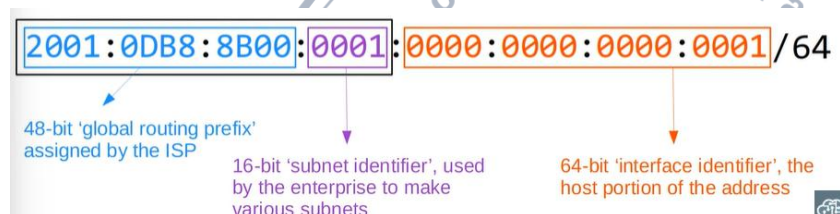
IPv6 Address Types

Global Unicast Addresses (Exam Topic 1.9.a)

- Public IPv6 addresses
- Used on the Internet
- Must be globally unique
- Must be registered

Originally defined as the 2000::/3 block (2000:: to 3FFF:FFFF:FFFF:FFFF:FFFF:FFFF:FFFF:FFFF).
Now defined as all addresses which aren't reserved for other purposes.

Global Unicast Address Structure



Components:

- Global routing prefix
- Subnet identifier
- Interface identifier

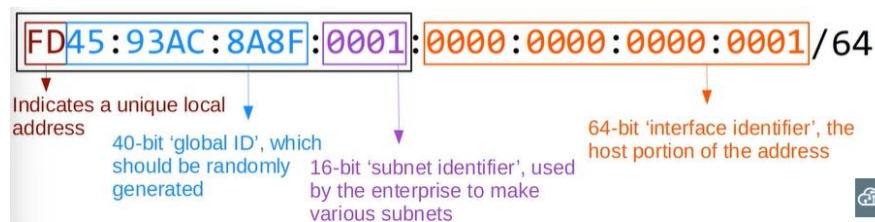
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Unique Local Addresses (Exam Topic 1.9.b)

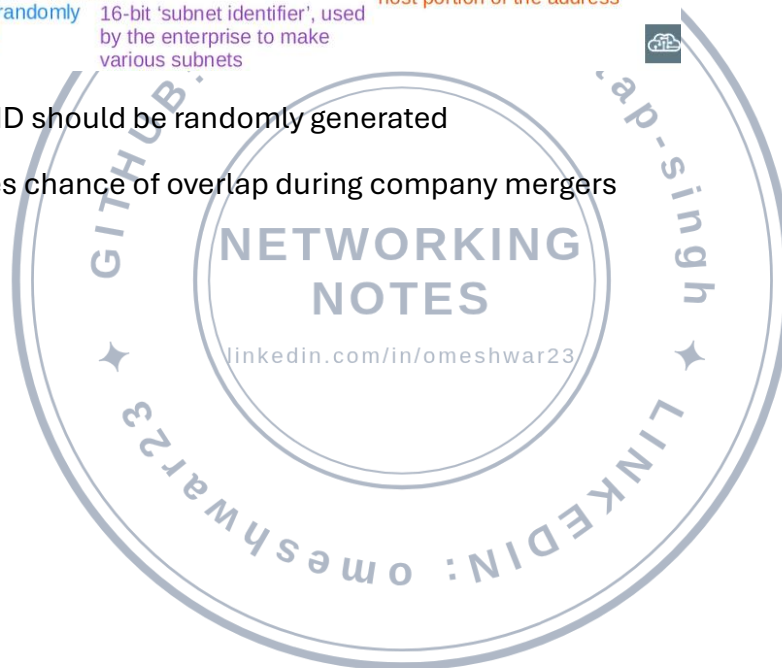
- Private IPv6 addresses
- Not routed over the Internet
- No registration required
- Similar to private IPv4 addresses

Uses the address block FC00::/7 (FC00:: to FDFF:FFFF:FFFF:FFFF:FFFF:FFFF:FFFF:FFFF)
However, a later update requires the 8th bit to be set to 1, so the first two digits must be FD.

Unique Local Address Structure



- Global ID should be randomly generated
- Reduces chance of overlap during company mergers



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Link-Local Addresses (Exam Topic 1.9.c)

- Automatically generated on IPv6-enabled interfaces

Uses the address block FE80::/10 (FE80:: to FEBF:FFFF:FFFF:FFFF:FFFF:FFFF:FFFF:FFFF)
However, the standard states that the 54 bits after FE80/10 should be all 0, so you won't see link local addresses beginning with FE9, FEA, or FEB. Only FE8.

The interface ID is generated using EUI-64 rules.

Link-Local Verification:

The interface ID is generated using EUI-64 rules.

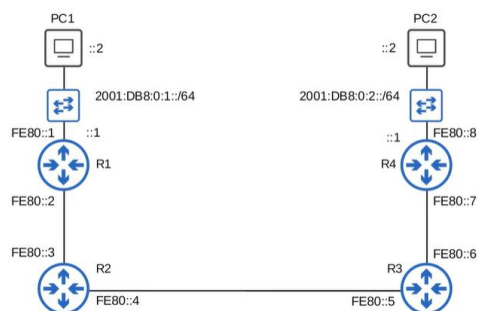
```
R1(config-if)#do show ipv6 interface brief
GigabitEthernet0/0    [up/up]
FE80::EF8:22FF:FE36:8500
2001:DB8::EF8:22FF:FE36:8500
GigabitEthernet0/1    [up/up]
FE80::EF8:22FF:FE36:8501
2001:DB8::0:1:EF8:22FF:FE36:8501
GigabitEthernet0/2    [up/up]
FE80::EF8:22FF:FE36:8502
2001:DB8::0:2:EF8:22FF:FE36:8502
GigabitEthernet0/3    [administratively down/down]
unassigned
```

Link-local and global unicast addresses share the same interface ID.

Purpose of Link-Local Addresses

- Used only within a single link
- Not routed between subnets
- Used for:
 - Routing protocol neighbour adjacencies (OSPFv3)
 - Next-hop addresses
 - Neighbour Discovery Protocol (NDP)

Link-Local Routing Example



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Packet Flow (Next-Hop Logic)

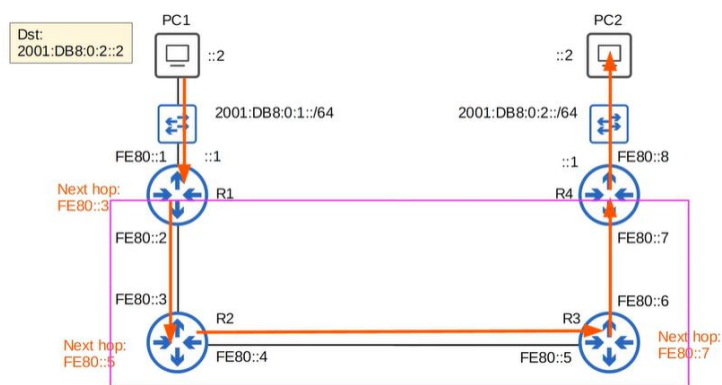
PC1 → R1 → R2 → R3 → R4 → PC2

Next hops:

- R1 → FE80::3
- R2 → FE80::5
- R3 → FE80::7

Routers forward packets using link-local next-hop addresses.

Routers cannot ping distant routers using link-local addresses.



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Multicast Addresses

Communication Types Review

- Unicast addresses are one-to-one
→ One source to one destination
- Broadcast addresses are one-to-all
→ One source to all destinations within the subnet
- Multicast addresses are one-to-many
→ One source to multiple destinations
→ Only hosts that joined the specific multicast group receive the traffic

Multicast

IPv6 uses the multicast range: FF00::/8

IPv6 uses range FF00::/8 for multicast. (FF00:: to FFFF:FFFF:FFFF:FFFF:FFFF:FFFF:FFFF:FFFF)

IPv6 does not use broadcast.

There is no broadcast address in IPv6.

Instead, IPv6 uses multicast to send traffic to all hosts in a subnet.

Important IPv6 Multicast Addresses

Purpose	IPv6 Address	IPv4 Address
All nodes/hosts (functions like broadcast)	FF02::1	224.0.0.1
All routers	FF02::2	224.0.0.2
All OSPF routers	FF02::5	224.0.0.5
All OSPF DRs/BDRs	FF02::6	224.0.0.6
All RIP routers	FF02::9	224.0.0.9
All EIGRP routers	FF02::A	224.0.0.10

Key observation:

- IPv4 and IPv6 multicast addresses use the same last digit
- Hexadecimal A = decimal 10

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All-Nodes Multicast Address

FF02::1

- Also called **all hosts**
- Functions like a broadcast
- Used to reach all hosts in the subnet

This is how IPv6 replaces broadcast behaviour.

IPv6 Multicast Scopes

IPv6 multicast addresses define scopes, which determine how far multicast traffic can travel.

The scope is identified by the 4th hexadecimal character.

1] Interface-Local Scope (FF01)

- Prefix: FF01
- Messages stay within the local device
- Do not leave the interface
- Also called node-local [linkedin.com/in/omeshwar23](https://www.linkedin.com/in/omeshwar23)

2] Link-Local Scope (FF02)

- Prefix: FF02
- Traffic stays within the local subnet
- Not routed between subnets
- Used by many controls' protocols

This is different from IPv6 link-local addresses (FE80::)

3] Site-Local Scope (FF05)

- Prefix: FF05
- Can be forwarded by routers
- Intended to stay within a single physical site
- Actual boundaries must be configured by the network engineer

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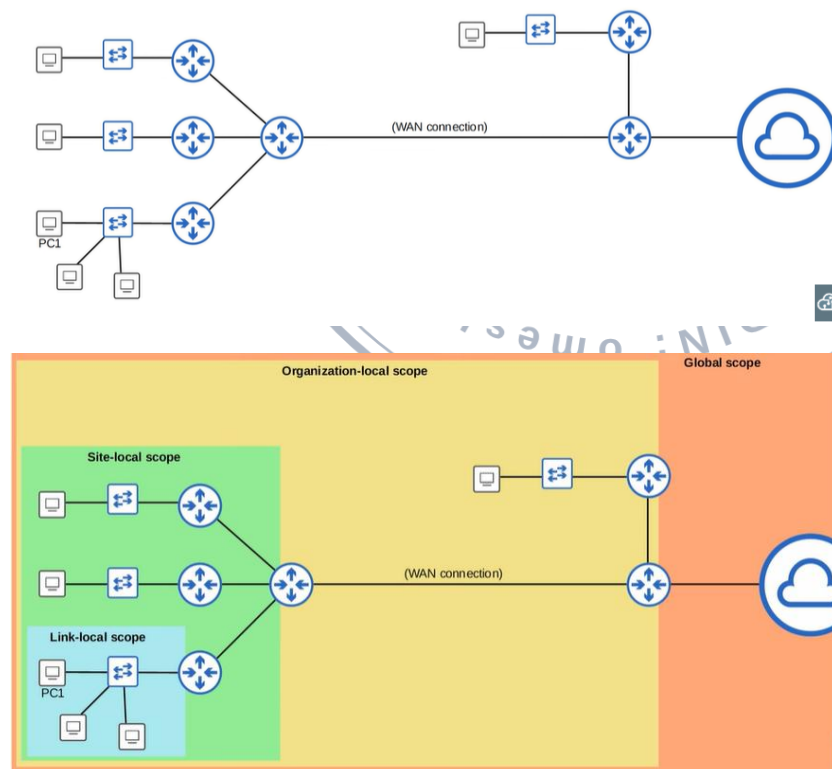
4] Organization-Local Scope (FF08)

- Prefix: FF08
- Wider than site-local
- Intended for the entire organization
- Boundaries are defined by configuration

5] Global Scope (FF0E)

- Prefix: FF0E
- Can be routed over the public Internet
- Defined as having no boundaries
- Still subject to network configuration

Multicast Scope Visualization



- Interface-local and link-local are clearly defined
- Other scopes require manual configuration
- Exact behaviour is beyond CCNA scope

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Joined Multicast Groups (Verification)

CLI Output:

```
R1#sh ipv6 int g0/0
GigabitEthernet0/0 is up, line protocol is up
IPv6 is enabled, link-local address is FE80::EF8:22FF:FE36:8500
No Virtual link-local address(es):
Global unicast address(es):
  2001:DB8::EF8:22FF:FE36:8500, subnet is 2001:DB8::/64 [EUI]
Joined group address(es):
  FF02::1
  FF02::2
  FF02::1:FF36:8500
MTU is 1500 bytes
ICMP error messages limited to one every 100 milliseconds
ICMP redirects are enabled
ICMP unreachable are sent
ND DAD is enabled, number of DAD attempts: 1
ND reachable time is 30000 milliseconds (using 30000)
ND advertised reachable time is 0 (unspecified)
ND advertised retransmit interval is 0 (unspecified)
ND router advertisements are sent every 200 seconds
ND router advertisements live for 1800 seconds
ND advertised default router preference is Medium
Hosts use stateless autoconfig for addresses.
```

Joined group address(es):

FF02::1

FF02::2

FF02::1:FF36:8500

Explanation:

- FF02::1 → All-nodes multicast
- FF02::2 → All-routers multicast
- The third group is a special multicast address (explained later)

R1 automatically joins:

- All-nodes group (it is a host on the subnet)
- All-routers group (it is a router)

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Anycast Addresses (Exam Topic 1.9.d)

Anycast is a new feature in IPv6.

- Anycast communication is one-to-one-of-many
- There are multiple possible destinations
- Traffic is delivered to one destination

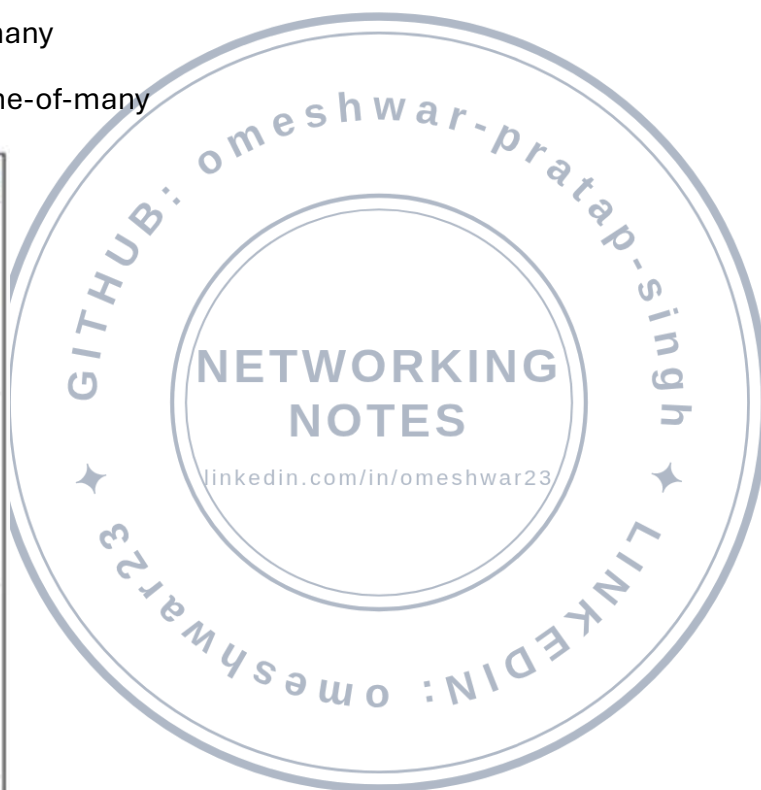
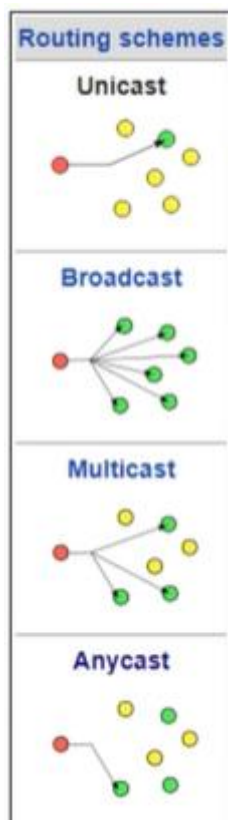
Anycast Concept Comparison

Unicast: 1 → 1

Broadcast: 1 → all

Multicast: 1 → many

Anycast: 1 → one-of-many



How Anycast Works

- Multiple routers are configured with the same IPv6 address
- Routers advertise the address using a routing protocol
- Traffic is forwarded to the nearest router
- “Nearest” means lowest routing protocol metric

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Anycast Address Range

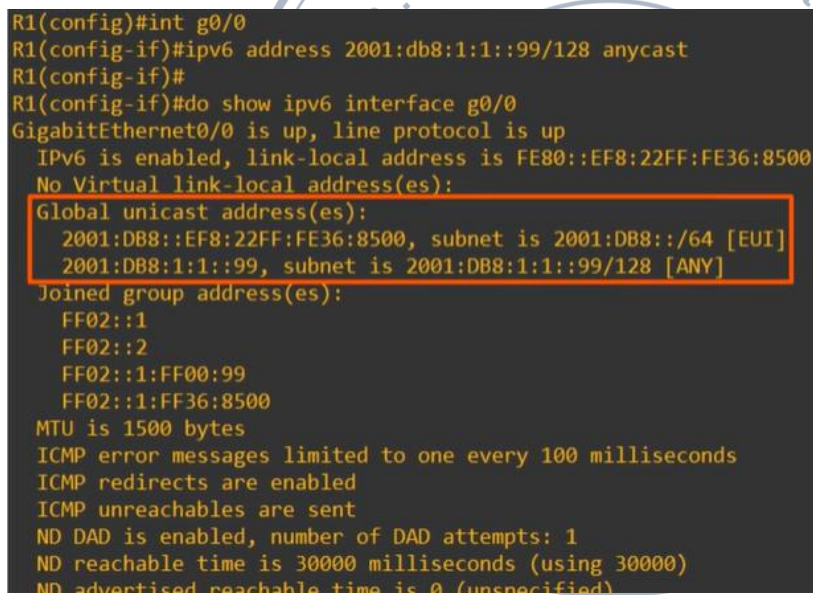
- There is no specific address range for anycast
- Use a normal unicast address:
 - Global unicast, or
 - Unique local
- Specify it as anycast

Anycast Configuration Example

R1(config-if)# ipv6 address 2001:db8:1:1::99/128 anycast

- /128 is like a /32 in IPv4
- A host route to this address will be advertised

Verification Output



```
R1(config)#int g0/0
R1(config-if)#ipv6 address 2001:db8:1:1::99/128 anycast
R1(config-if)#
R1(config-if)#do show ipv6 interface g0/0
GigabitEthernet0/0 is up, line protocol is up
IPv6 is enabled, link-local address is FE80::EF8:22FF:FE36:8500
No Virtual link-local address(es):
Global unicast address(es):
  2001:DB8::EF8:22FF:FE36:8500, subnet is 2001:DB8::/64 [EUI]
  2001:DB8:1:1::99, subnet is 2001:DB8:1:1::99/128 [ANY]
Joined group address(es):
  FF02::1
  FF02::2
  FF02::1:FF00:99
  FF02::1:FF36:8500
MTU is 1500 bytes
ICMP error messages limited to one every 100 milliseconds
ICMP redirects are enabled
ICMP unreachable are sent
ND DAD is enabled, number of DAD attempts: 1
ND reachable time is 30000 milliseconds (using 30000)
ND advertised reachable time is 0 (unspecified)
```

Global unicast address(es):

2001:DB8::EF8:22FF:FE36:8500 [EUI]

2001:DB8:1:1::99/128 [ANY]

- Anycast addresses still appear under **global unicast**
- [ANY] indicates it is an anycast address

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Other IPv6 Address

- `::` = The *unspecified* IPv6 address
 - ↳ Can be used when a device doesn't yet know its IPv6 address.
 - ↳ IPv6 default routes are configured to `::/0`
 - ↳ IPv4 equivalent: `0.0.0.0`
- `::1` = The loopback address
 - ↳ Used to test the protocol stack on the local device.
 - ↳ Messages sent to this address are processed within the local device, but not sent to other devices.
 - ↳ IPv4 equivalent: `127.0.0.0/8` address range

Summary of Topics Covered

- IPv6 multicast addresses and their purpose
- IPv6 does not use broadcast
- Common IPv6 multicast addresses
- Multicast scopes and their meanings
- Anycast addresses and how they work
- Unspecified IPv6 address (`::`)
- IPv6 loopback address (`::1`)

This lesson contains many concepts.

Review each topic one by one as needed.